

Seunghwan Jang

Ph.D. Student, School of Electrical & Electronic Engineering, Nanyang Technological University

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Education

Nanyang Technological University (NTU)

Ph.D. Student, School of Electrical & Electronic Engineering

Singapore

Aug. 2026 – Present

- Advisor: Prof. Wei Xiao

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Electrical Engineering

Daejeon, Korea

Sep. 2024 – Aug. 2026

- Advisor: Prof. SooJean Han

Kyung Hee University

B.E. in Computer Science & Engineering

Yongin, Korea

Mar. 2019 – Aug. 2024

- GPA: 4.02/4.3 (**Ranked 1st/97**), College of Software Convergence
- Advisor: Prof. Hui Yong Kim

Publications

Conference

- [1] **Seunghwan Jang**[†], Wonje Jeung, and SooJean Han, “Systematic Hazard Sampling: Minimal-Variance Inference for Discrete Diffusion and Flow Models,” *Conference on Neural Information Processing Systems (NeurIPS) 2026*. (Earlier version: “Stratified Hazard Sampling,” *ICLR 2026 ReALM-GEN Workshop*)
- [2] **Seunghwan Jang**[†], Jeongyong Yang, Siddharth Ancha, and SooJean Han[†], “Safe Streaming Flow Planning by Aligning Sampling Dynamics with Execution Dynamics,” *Conference on Robot Learning (CoRL) 2026*. [[OpenReview](#)] (Earlier version: *ICLR 2026 Workshop on World Models*)
- [3] Jeongyong Yang^{*}, **Seunghwan Jang**^{*†}, and SooJean Han[†], “SafeFlowMatcher: Safe and Fast Planning using Flow Matching with Control Barrier Functions,” *International Conference on Learning Representations (ICLR) 2026*. [[Open-Review](#)]

(* Equal contribution; † Corresponding author)

Workshop

- [4] **Seunghwan Jang** and SooJean Han, “Comparison of Compression Codec Algorithms to Image Datasets in 3D Gaussian Splatting Models,” *Workshop on Image Processing and Image Understanding (IPIU)*, Poster, Feb. 2025. (in Korean)

Research Experience

School of Electrical & Electronic Engineering, NTU

Ph.D. Student (Advisor: Prof. Wei Xiao); Research Assistant, Jun.–Aug. 2026

Singapore

Jun. 2026 – Present

- Safety certification for learning-based and generative models.

Autonomous Control for Stochastic Systems Lab, KAIST

M.S. Student (Advisor: Prof. SooJean Han)

Daejeon, Korea

Sep. 2024 – Aug. 2026

- Safe generative motion planning: flow-matching and diffusion trajectory generation coupled with CBFs safety filtering.
- Inference-time theory for discrete diffusion: minimal-variance event scheduling.

Visual Media Lab, Kyung Hee University

Research Student (Advisor: Prof. Hui Yong Kim)

Yongin, Korea

Aug. 2022 – Aug. 2024

- Built and evaluated MPEG FCM/FCVCM pipelines combining conventional codecs (VVC, HEVC) with neural compression, focusing on rate-accuracy trade-offs and reproducible experiments.
- 3 first-author domestic publications; 10 MPEG FCM/FCVCM standardization contributions; 1 Korean patent.

Electronics and Telecommunications Research Institute (ETRI)

Research Intern

Daejeon, Korea
Jul. 2023 – Aug. 2023

- Supported MPEG standardization efforts by conducting validation experiments and analyzing performance data for FCM/FCVCM.

Projects

Fast, Accurate, and Efficient Situational Awareness using Visual Sensor Networks

Jul. 2025 – Aug. 2026

Air Force Office of Scientific Research (AFOSR), United States

- Developing a 3D Gaussian Splatting-based visual sensor networks system.

Honors & Awards

Graduated first in class (College of Software Convergence), Kyung Hee University (1st/97)

Aug. 2024

Superiority Scholarship, Kyung Hee University

Feb. 2024

Superiority Scholarship (Half tuition), Kyung Hee University

Aug. 2019

Skills

Programming: Python, C/C++

Robotics/Simulation: ROS, Gazebo, MuJoCo, Gym

Machine Learning: PyTorch

Tools: Git, Linux, Docker

Peer Review Experience

- Conference on Robot Learning (CoRL), 2026
- IEEE Robotics and Automation Letters (RA-L), 2025, 2026

Languages

Korean: Native

English: Professional working proficiency

Japanese: Professional working proficiency